

INPUT & OUTPUT DEVICES PRACTICE SHEET

NIMCET TARGET PAPER • ADVANCED CONCEPTS (HARD - THEORETICAL)

ResourceOS

1. In Memory-Mapped I/O configuration, which CPU instruction set feature allows bit-manipulation directly on I/O device registers without loading into CPU accumulators first?

- (A) Special IN and OUT port instructions
- (B) Full memory-referencing instructions (e.g., OR, AND, MOV)
- (C) Dedicated bus-arbitration hardware macros
- (D) Vector table indirect addressing mode

2. Why is an Isolated I/O architecture advantageous when designing systems with complex memory address decoders?

- (A) It reduces hardware wiring because memory and I/O share identical control signal lines.
- (B) Memory address decoding logic does not need to accommodate I/O port address ranges.
- (C) It allows DMA controllers to perform memory-to-I/O transfers without bus arbitration.
- (D) It automatically eliminates the need for software status polling loops.

3. What distinguishes "Burst Mode" DMA from "Cycle Stealing" DMA transfers?

- (A) Burst mode transfers data byte-by-byte only when CPU is idle in T-states.
- (B) Burst mode retains system bus ownership continuously for a complete block transfer before relinquishing it back to CPU.
- (C) Cycle stealing forces CPU hardware reset on every memory word read.
- (D) Burst mode cannot operate with high-speed secondary disk drives.

4. In a Daisy-Chained hardware interrupt structure, what determines the priority level assigned to a requesting device?

- (A) The numerical value of the interrupt vector sent over the data bus
- (B) The physical proximity of the device along the Interrupt Acknowledge (INTA) signal chain relative to CPU
- (C) The clock frequency of the peripheral interface controller
- (D) The value written into the CPU control register by the operating system kernel

5. What is the major limitation of Daisy-Chain hardware priority interrupt arbitration schemes?

- (A) High hardware cost due to requiring dynamic priority encoder chips
- (B) Vulnerability to starvation of lower-priority devices located further down the chain
- (C) Inability to support non-vectored interrupt service routines
- (D) Mandatory requirement for dual-port RAM buffers

6. How does "Transparent DMA" (Hidden DMA) achieve zero CPU cycle loss during high-speed data transfers?

- (A) By stealing CPU instruction fetch cycles explicitly using HOLD hardware interrupts
- (B) By accessing system memory only during CPU internal execution phases (T-states) when the system bus is unutilized
- (C) By operating over a secondary dedicated instruction bus completely isolated from RAM
- (D) By executing data transfers entirely inside the CPU cache memory

7. What critical role is played by an Interrupt Vector Table (IVT) during Vectored Interrupt execution?

- (A) Storing pending interrupt requests in a FIFO priority queue
- (B) Mapping an interrupt vector identifier supplied by a device to the memory address of its specific ISR
- (C) Preserving CPU accumulator and flag registers during context switching
- (D) Arbitrating bus grant conflicts between DMA and I/O processors

8. In asynchronous data transfer, what fundamental flaw exists in a Simple Handshaking mechanism using only a single "Strobe" signal?

- (A) The transmitter has no confirmation whether the receiver has actually accepted data from the bus.
- (B) Data cannot be transmitted in parallel across 8-bit or 16-bit data lines.
- (C) The CPU must run in fully synchronous clock-lock step with the peripheral.
- (D) Parity bits cannot be appended to the transmitted character byte.

9. Why is two-wire full-handshake control (Data Ready & Data Accepted) superior to source-initiated strobe timing?

- (A) It eliminates the need for any data bus line interconnections.
- (B) It allows units operating at widely varying data transfer rates to communicate reliably without data loss.
- (C) It removes the necessity of saving context state on the CPU stack.
- (D) It allows Direct Memory Access without acquiring system bus control.

10. In a Random-Scan (Vector) CRT display, how are visual image primitives rendered onto the phosphorescent screen?

- (A) The electron beam sweeps in a rigid top-to-bottom, left-to-right grid pattern.
- (B) The electron beam is directed only along the exact line path coordinates specified by display commands.
- (C) Image frames are stored as a two-dimensional grid of discrete raster pixel intensity maps.
- (D) The screen employs light-emitting diodes that illuminate independently without electron deflection.

11. Why do Random-Scan (Vector) displays produce smoother lines and curves compared to Raster-Scan displays?

- (A) Vector displays utilize higher color-depth lookup tables.
- (B) Vector lines are drawn directly as continuous physical line segments without pixel staircase jaggedness (aliasing).
- (C) Vector displays operate at refresh frequencies exceeding 240 Hz.
- (D) Vector displays eliminate horizontal and vertical retrace timing delays.

12. What structural architectural component distinguishes an I/O Processor (I/O Channel) from a standard DMA Controller?

- (A) An I/O Processor can decode and execute its own specialized instruction set stored directly in main memory.
- (B) A DMA Controller can execute high-level operating system system calls autonomously.
- (C) An I/O Processor does not require system bus control signals to transfer data to RAM.
- (D) A DMA Controller contains internal Arithmetic Logic Units (ALU) for image processing.

13. What occurs during the "Horizontal Retrace" interval in CRT raster scanning graphics systems?

- (A) The electron beam is turned off (blanked) as it moves back from the end of a scan line to the start of the next.
- (B) The frame buffer memory contents are flushed and re-initialized from system RAM.
- (C) The electron beam moves from the bottom-right corner to the top-left corner of the screen.
- (D) The display controller re-synchronizes the hardware priority encoder logic.

14. In software-driven polling (Programmed I/O), what factor most directly degrades overall system throughput?

- (A) Memory bus lock delays caused by DMA burst transmissions
- (B) CPU processing cycles consumed executing status-check instructions repeatedly while peripheral is idle
- (C) Excessive context switches saving registers on the system stack
- (D) Hardware logic delays along the priority daisy-chain signal lines

15. What is the functional purpose of a Priority Encoder module in parallel hardware interrupt controllers?

- (A) It translates active interrupt request lines into a binary code corresponding to the highest priority active request.
- (B) It converts serial data bits received from peripherals into parallel bytes for CPU registers.
- (C) It prevents DMA controllers from overriding CPU memory fetch operations.
- (D) It decodes memory address offsets into physical chip selection signals.

16. Which statement accurately describes the operation of a Non-Maskable Interrupt (NMI)?

- (A) It can be disabled by clearing the interrupt enable (IF) flag in the CPU status register.
- (B) It has high hardware priority and must be serviced immediately, bypassing internal software mask flags.
- (C) It is generated strictly by software execution of trap or breakpoint instructions.
- (D) It can only be issued by standard low-speed serial peripherals like keyboards.

17. In an impact Dot Matrix printer, how is variation in font weight or print density produced physically?

- (A) By adjusting the chemical thermal composition of the output paper
- (B) By firing print pins multiple times over overlapping dot passes (multipass printing)
- (C) By altering the voltage applied to electrostatic toner drums
- (D) By varying the liquid ink jet droplet velocity from piezoelectric nozzles

18. What key functional mechanism enables a Laser Printer to transfer powdered toner onto physical paper?

- (A) High-pressure impact force exerted by mechanical hammers
- (B) Electrostatic charge pattern created on a photoconductive drum surface by a laser beam
- (C) Thermal heating of heat-sensitive chemical dyes impregnated in paper
- (D) Continuous deflection of pressurized liquid ink streams via electrostatic plates

19. How does a Resistive Touchscreen differ fundamentally in touch detection mechanism from a Capacitive Touchscreen?

- (A) Resistive screens rely on physical pressure causing two conductive layers to make electrical contact.
- (B) Resistive screens measure local distortion of an electrostatic field caused by conductive human skin.
- (C) Resistive screens utilize surface acoustic sound wave reflection grids.
- (D) Resistive screens require active infrared optical sensors along screen borders.

20. Why must the CPU save the Program Counter (PC) and Status Register (Flags) onto the stack before executing an Interrupt Service Routine (ISR)?

- (A) To allow the DMA controller to modify instruction addresses during interrupt processing
- (B) To ensure the interrupted program can resume execution accurately from the exact point of interruption
- (C) To prevent higher-priority hardware interrupts from firing while the ISR executes
- (D) To clear memory space for storing peripheral input data buffers

21. In computer display technology, what is "Flicker" primarily caused by?

- (A) Low spatial resolution (small pixel count) across the display matrix
- (B) Refresh rate falling below the human eye's critical fusion frequency (phosphor decay rate)
- (C) Insufficient frame buffer memory color bit depth
- (D) High dot pitch distance between neighboring subpixels

22. What happens when a Software Interrupt instruction (such as INT n or TRAP) is executed by the CPU?

- (A) An external hardware pin receives an active low signal from a peripheral controller. (B) The CPU synchronously suspends current execution and branches to an ISR corresponding to vector number n.
- (C) System bus arbitration immediately transfers control to the DMA channel. (D) The CPU enters a halt state until an external hardware reset occurs.

23. What is the principal disadvantage of Memory-Mapped I/O compared to Isolated (Port-Mapped) I/O architecture?

- (A) Memory-Mapped I/O requires separate CPU control lines for I/O READ and I/O WRITE operations. (B) I/O device registers consume addresses that would otherwise be available for physical system RAM.
- (C) It prevents the use of powerful memory-referencing instructions for I/O operations. (D) Data transfers must pass exclusively through a Direct Memory Access (DMA) channel.

24. In centralized bus arbitration with independent request and grant lines, how is arbitration delay affected relative to Daisy Chaining?

- (A) Delay is higher because arbitration signals propagate sequentially through every device. (B) Delay is constant and independent of the number of devices because requests reach the central arbiter directly in parallel.
- (C) Delay depends strictly on the physical lengths of data lines on the main motherboard. (D) Delay increases exponentially as priority levels are dynamically rotated.

25. In modern high-speed peripheral interfacing, what primary operational advantage does Serial Transmission hold over traditional Parallel Transmission for long distances?

- (A) Parallel transmission suffers from signal skew (differing arrival times) and crosstalk between parallel wires at high clock rates. (B) Serial transmission transmits full multi-byte words simultaneously over a single optical channel.
- (C) Parallel transmission requires complex asynchronous handshaking protocols not needed in serial. (D) Serial transmission completely eliminates the need for frame synchronization bits.

COMPLETE ANSWER KEY (QUESTIONS 1 - 25)

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(B)	6	(B)	11	(B)	16	(B)	21	(B)
2	(B)	7	(B)	12	(A)	17	(B)	22	(B)
3	(B)	8	(A)	13	(A)	18	(B)	23	(B)
4	(B)	9	(B)	14	(B)	19	(A)	24	(B)
5	(B)	10	(B)	15	(A)	20	(B)	25	(A)